

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-blind, Placebo-controlled, Multicenter Basket Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab in Patients With Anti-N-methyl-D-aspartic Acid Receptor (NMDAR) or Anti-leucine-rich Glioma-inactivated 1 (LGI1) Encephalitis **Short Title/Acronym:** CIELO **Protocol Number:** WN43174 **NCT Number:** NCT05503264 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Satralizumab **Phase of Development:** Phase III **Trial Status:** Recruiting **Medical Monitor:** Hoffmann-La Roche **Clinical Operations Lead**

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of satralizumab compared with placebo in participants with NMDAR-immunoglobulin G antibody-positive (IgG+) or LGI1-IgG+ autoimmune encephalitis (AIE). **Secondary Objectives:** To evaluate the safety, pharmacodynamics (PD), pharmacokinetics (PK), and exploratory efficacy endpoints including cognitive function, time to rescue therapy, and seizure cessation.

2.2 Background & Rationale To address the critical unmet medical need in patients with autoimmune encephalitis (AIE), specifically NMDAR and LGI1 AIE. AIE is a rare disease caused by brain inflammation where autoantibodies mistakenly attack healthy tissues. Currently, there are no approved therapies for AIE. Interleukin-6 (IL-6) signaling plays a significant role in the pathophysiology of AIE, and satralizumab (an IL-6 receptor inhibitor) has the potential to be an evidence-based disease-modifying treatment.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Basket study) **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 152 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 27-Sep-2022 **Estimated Primary Completion:** 16-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 12 years with a diagnosis of probable or definite NMDAR encephalitis for the NMDAR AIE Cohort. 2. Age ≥ 18 years with a diagnosis of LGI1 encephalitis for the LGI1 AIE Cohort. 3. Reasonable exclusion of tumor or malignancy before baseline visit (randomization). 4. Onset of AIE symptoms ≤ 9 months before randomization. 5. Meet the definition of "New Onset" or "Incomplete Responder" AIE. 6. For women of childbearing potential: agreement to remain abstinent or use adequate contraception during the treatment period and for at least 3 months after the final dose. 7. For participants enrolled in the extended China enrollment phase: current residents of mainland China, Hong Kong, or Taiwan, and of Chinese ancestry.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding, or intending to become pregnant during the study or within 3 months after the final dose. 2. Any untreated teratoma or thymoma at baseline visit (randomization). 3. History of carcinoma or malignancy, unless deemed cured by adequate treatment with no evidence of recurrence for ≥ 5 years before screening. 4. For participants with NMDAR AIE, history of negative anti-NMDAR antibody in cerebrospinal fluid (CSF) using a cell-based assay within 9 months of symptom onset. 5. Historically known positivity to an intracellular antigen with high cancer association, glutamate decarboxylase 65 (GAD-65), or cell surface neuronal antibodies other than NMDAR/LGI1 in the absence of NMDAR/LGI1 positivity. 6. Confirmed paraneoplastic encephalitis or central/peripheral nervous system demyelinating disease. 7. History of herpes simplex virus encephalitis in the previous 24 weeks or alternative causes of associated symptoms. 8. Previous or concurrent treatment with IL-6 inhibitors, alemtuzumab, total body irradiation, bone marrow transplantation, CD19 antibodies, complement inhibitors, neonatal Fc receptor antagonists, T-cell depleting therapies, or oral cyclophosphamide within 1 year prior to baseline. 9. Concurrent use of more than one immunosuppressive therapy (IST) as background therapy, or contraindication to all rescue treatments (rituximab, IVIG, high-dose corticosteroids, or IV cyclophosphamide). 10. Active, recurrent, or serious infections, including HIV, positive HBV/HCV, or evidence of latent/active tuberculosis (TB). 11. History of suicide attempt within 3 years prior to screening (except if clearly associated with the acute phase of LGI1 or NMDAR encephalitis). 12. Receipt of live or live-attenuated vaccine within 6 weeks prior to baseline, or history of blood/plasma/platelet donation within 90 days.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Satralizumab or matching placebo administered at Weeks 0, 2, 4, and every 4 weeks thereafter. **Route of Administration:** Subcutaneous (SC) injection

Duration of Treatment: 52-week primary treatment period (Part 1), followed by an optional extension period (Part 2) that may last up to 5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Participants may continue to receive background immunosuppressive therapy, symptomatic treatment, and rescue therapy throughout the study. **Prohibited:** Receipt of live or live-attenuated vaccine within 6 weeks prior to baseline.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 1 month before Day 0	Informed Consent, Medical History, Confirmation of AIE diagnosis, Exclusion of malignancies
2	Day 0	Randomization, First Dose of Satralizumab/Placebo
3	Weeks 2, 4, and every 4 weeks	Subcutaneous Injections, Efficacy/Safety Assessments, Cognitive Tests
4	Week 52	Final Primary Period Assessments, Option to enter extension phase

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Part 1 (NMDAR AIE Cohort): Proportion of participants with a modified Rankin Scale (mRS) score improvement ≥ 1 from baseline and no use of rescue therapy at Week 24. **2. Part 1 (LGI1 AIE Cohort):** Proportion of participants with an mRS score improvement ≥ 1 from baseline and no use of rescue therapy at Week 52. **3. Part 2:** Percentage of participants with Adverse Events (AEs).

7.2 Secondary & Exploratory Endpoints 1. Time to mRS score improvement ≥ 1 from baseline without the use of rescue therapy. 2. Time to rescue therapy. 3. Proportion of participants with sustained seizure cessation (maintained until Week 24) and no use of rescue therapy. 4. Change in Clinical Assessment Scale of Autoimmune Encephalitis (CASE) score. 5. Cognitive function changes assessed via Montreal Cognitive Assessment and Rey Auditory Verbal Learning Test (RAVLT).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection during every 4-week clinic visit, including assessment of suicidal ideation/behavior via the C-SSRS. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to oversee patient safety and trial integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Target enrollment of $N=152$ is powered to detect statistically significant improvements in the mRS score between the satralizumab and placebo arms. **Handling of Missing Data:** Participants requiring rescue therapy or dropping out before their primary evaluation point will be treated as non-responders for the primary endpoint analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol adherence and data integrity. **Audit Trail:** Standardized electronic Case Report Forms (eCRF) with detailed audit trails and data clarification processes.

11. Study Identifiers & Governance

Primary ID: WN43174 **Registry Number:** NCT05503264 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** CIELO **Official Regulatory Title:** A Phase III, Randomized, Double-blind, Placebo-controlled, Multicenter Basket Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab in Patients With Anti-N-methyl-D-aspartic Acid Receptor (NMDAR) or Anti-leucine-rich Glioma-inactivated 1 (LGI1) Encephalitis

12. Clinical Framework (Autoimmune Encephalitis Specific)

Primary Condition: NMDAR Autoimmune Encephalitis, LGI1 Autoimmune Encephalitis **Diagnosis Threshold:** Confirmed NMDAR-IgG+ or LGI1-IgG+ with symptom onset ≤ 9 months **Medical Methodology:** Interleukin-6 (IL-6) receptor inhibition **Optimization Target:** Improvement in modified Rankin Scale (mRS) score without the need for rescue therapies

13. Study Procedures & Methodology

Management Pathway: Subcutaneous biologic administration alongside background symptomatic/immunosuppressive treatment **Education Protocol (HCP):** Training on specialized clinical evaluations including the mRS, CASE, and cognitive assessments (RAVLT). **Education Protocol (Patient):** Comprehensive informed

consent, treatment compliance education, and symptom tracking.
Quality Audit Mechanism: Centralized review of primary endpoint scoring to ensure inter-rater reliability.

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (primary treatment period), with an optional extension up to 5 years **Onsite Visit Cadence:** Weeks 0, 2, 4, and every 4 weeks thereafter **Remote Monitoring Frequency:** Inter-visit monitoring as clinically indicated. **Checklist Review Interval:** Every 4 weeks (Q4W) concurrent with dosing

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					